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Basic Configuration List for All Devices

enable

This accept command is the device enabled position.

config terminal

This is a global configuration mode prompt on the device

hostname {xx}

This command serves to name the device with {xx}

To configure an IP address on interface

To enter the interface use:

interface {interface-name}

Then, configure the IP address and subnet mask using:

ip address {subnet-mask} ip-address

To save the configuration:

Use the command: write memory

Basic Varification commands

Show running config: Display the current configuration

Show ip interfaces verifies that ACL is applied to the specified interface.

Single area OSPFv2

Definition of OSPF: OSPF achieves convergence through the exchange of link-state information, allowing routers to build a consistent view of the network topology. The use of DBD, LSR, LSU, and LSAck packets facilitates this process.

CAMMAND	DESCRIPTION
R1(config)# router ospf{process id}	Enters OSPF routing protocol configuration mode for the specified process ID.

R1(config-router)# router-id {x.x.x.x}	Assigns a unique OSPF router ID in the format of an IP address. This ID must be unique in the OSPF domain.
R1(config-router)# network {network-address} {wildcard-mask} area {area-id}	Defines which interfaces will participate in OSPF. The network address and wildcard mask specify the interfaces in the given area.
R1(config-if)# ip ospf cost {value}	Manually sets the cost of the OSPF route for the specified interface. Lower values indicate better routes.
R1# show ip ospf neighbor	Displays the OSPF neighbors and their states, useful for troubleshooting neighbor relationships.
R1# show ip route ospf	Displays the OSPF routes in the routing table
R1# show ip ospf	Provides general OSPF information, including router ID and area details

DR and BDR role examination IN OSPF

Definition of DR: the Designated Router (DR) is a router in an OSPF multi-access network responsible for generating and distributing link-state advertisements (LSAs) to other routers, thereby minimizing network traffic. It is elected based on the highest OSPF priority value configured on its interface, with ties broken by the router ID.

Definition of BDR: The Backup Designated Router (BDR) is a router that serves as a standby to the DR in an OSPF multi-access network. If the DR fails, the BDR takes over its responsibilities without initiating a new election. It is elected using the same criteria as the DR: highest OSPF priority and router ID in case of ties.

CAMMAND	Description
R1# show ip ospf neighbor	Displays the OSPF neighbors and their current roles (DR, BDR, or DROther).
R1# show ip ospf interface	Provides information on OSPF interfaces, including the DR and BDR for each network segment.
R1(config-if)# ip ospf priority {value}	onfigures the OSPF interface priority to influence the DR/BDR election. A higher value increases the chance of becoming DR.
R1(config-if)# no ip ospf priority	Resets the OSPF priority to the default value (1).
R1# clear ip ospf process	Restarts the OSPF process, which can trigger a new DR/BDR election based on the current priorities.

NAT configuration cammands:

Definition of NAT: **Network Address Translation (NAT)** is a technique used in computer networking that allows multiple devices on a local network to share a single public IP address when accessing external networks, such as the internet. NAT translates the private IP addresses of devices within a local network to a public IP address, allowing them to communicate with external networks while maintaining the privacy and security of the internal IP addresses

Commands	Discription
access-list 1 permit {starting address}, {ending address} and {sunet mask}	Create an access list to permit internal IPs for NAT.
ip nat inside source list 1 pool PUBLIC_ACCESS	Associate the access list with the NAT pool.
show ip nat translations	Display current NAT translations.
show ip nat statistics	show NAT statistics and usage information.